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Aluminum Ultra-Thin Films Grown by Physical Vapor Deposition for Solar Cell Electric Nanocontacts

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As aluminum film thickness and feature size fall into nanoscale, drastic and abrupt changes in optical and electric material properties occur. Reduction of film thickness beyond a critical thickness results in island formation and local nanostructure variation. Sputter-deposited Al thin films on Si wafers were studied and the structure of grown nanofeatures is characterized. Aluminum island nanostructure formation has been studied for a variety of films with reference to critical thickness. Uniformly distributed Al nanoparticles with a diameter of 10 to 30 nm were formed on the silicon surface for an aimed average film thickness of 15nm. Transitions of resistivity, reflectance, and nanofeature size were observed for film thickness around 40 nm. Equally important, these properties appeared well correlated with the transformation of film nanostructure as the Al deposition proceeds.

Introduction

Thanks to the compatibility of aluminum with silicon, Al films have been widely used for electronic and photovoltaic devices. The importance of Al is even higher for photovoltaics, not only for its low cost but also for it easily alloys with silicon and have useful electronic properties for making Al-Si junctions, which can be ohmic or Schottky, depending on the used deposition process and the substrate parameters. Al is also used for doping, making optical windows when oxidized,…. Traditionally Al has been used for rear electrodes in silicon solar cells, in the meantime, when thermally cycled adequately it creates a built-in back surface electric field (BSF) as a result of the formation of a pp^+ junction. Additionally, Al has good gettering ability of grown-in bulk impurities (1), which can be technologically useful.

Besides, Al alloys with many other elements which enabled producing a wide range of direct bandgap semiconductors such as AlP, AlAs,… for optoelectronics and photovoltaics.

Interest in new potential applications for modern photovoltaic devices has lately grown with third generation solar cells. For instance, the scattering on Al nanostructure formed on top of the solar cells can improve light trapping. Recent research on plasmonic photovoltaics has shown that metallic nanoparticles can direct light rays into solar cell active zone. Through excitation of the surface plasmons and the collective oscillations of

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conduction electrons in metals, the incident light may interact with the nanostructures in a resonant manner and scatter the light over a cross-sectional area larger than its geometric cross-section (2). Thus, for plasmonic nanostructures located close to the cell surface, a significant part of the scattered sunlight can be coupled with the trapped waveguide modes of the photoactive layer (3, 4). The scattering by the nanoparticles increase the optical path in the cell absorber. Hence it improves the optical absorption and produces larger photocurrent. It is possible to obtain a total light absorption higher than for conventional solar cells.

Light absorption efficiency of solar cells has been modeled for various materials and size of nanoparticles, in particular for silver and aluminum nanoparticles (5). Silver particles induce optical resonances in the visible part of the spectrum that are better at focusing the light into the solar cell. Conversely, optical resonance can cause light absorption by nanoparticles, which reduces solar cell efficiency. For silver, this resonance is indeed right in the key part of the solar spectrum, leading to a sizable light loss, whereas for aluminum, these resonances are outside the important part of the solar spectrum. Therefore, properties of light scattering on Al nanoparticles are compelling in comparison with silver nanoparticles. It has been found, indeed, that Al nanoparticles enhance light trapping in thin-film solar cells more than nanoparticles made of other metals. As a FCC metal, bulk Al has a band structure similar in shape to free-electron bands (6), which leads to a high reflectivity throughout the visible spectrum. Noteworthy, an absorption band appears when dealing with film thickness around 80 nm.

Aluminum has a valence of three electrons, supporting the high conductivity and reflectivity of the material (7). The unique optical and electrical properties of Al nanoparticles make it a promising material in the applications of photovoltaic and optoelectronic devices. Therefore, the understanding of the optical and electric characteristics of Al ultra-thin layers as a function of critical thickness becomes extremely important, particularly in the nanometer scale. In this paper, we report on the fabrication of ultra-thin films and Al nanoparticles by sputter-deposition. We discuss data on the reflectance and resistivity as a function of thickness chosen to be below the critical thickness, where discontinuous Al films are formed.

Experimental

In preparation of deposition of our Al ultra-thin film samples, 2" polished Si (111) wafers were subjected to a JTB-111 cleaning process, and dried by flowing nitrogen. Al at a purity of 99.99% was then deposited on top of the Si wafers with various “aimed thicknesses” using magnetron sputtering technique. The base pressure in the reaction chamber was 2 x 10^{-6} Torr, and the sputtering was performed at RT at 43 mTorr with an argon flow of 30 sccm. We had loaded the sputtering system to its maximum load of 12 wafers. Two samples were taken out from the reaction chamber when each aimed thickness was reached. In order to avoid the oxidation of the Al film for the remaining samples, N_2 gas was continuously flown into the chamber during sample unloading. FE-SEM was carried out to investigate the nanostructure of Al thin film. The thickness of Al film was measured across a step by a Dektak II profilometer, and the sheet resistance was evaluated using 4-point probe Magnetron M-700. Table I gives the aimed thickness versus actually obtained Al film thickness. Reflectance was measured using UV-Vis-NIR fiber probe based spectrometer.

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Table I: PVD grown Al film on silicon.

Sample No.	1	2	3	4	5	6	7	8
Aimed thickness (nm)	10	20	30	40	60	80	100	120

Results and Discussion

1) Structural Properties of Aluminum Ultra-Thin Film

Table II provides film thickness obtained using Dektak II. The given thicknesses are average values of the depth profile obtained by measuring a step made by masking a portion of the silicon substrate from the sputtered material. The given standard deviation in fact includes the nanostructure at the surface of the film and the Si substrate initial roughness as well as all source of measurement errors (e.g., sample vibration, tip jumps related to feature and tip curvatures and scanning speed, etc.). To avoid effects of the unevenness of the surface, the integration was limited to a lateral range of 150 µm.
One can notice that the standard deviation of film thickness for Sample 1 is higher than that of Sample 8, which has a thicker film. The peak shows a maximum variation of the surface nanofeature height for Sample 3, which is about three times that of the thickest sample, i.e., Sample 8.

Table II : Measured Al film thickness corresponding with the standard deviation.

Sample No.	1	2	3	4	5	6	7	8
Measured thickness (nm)	15.4	25.6	39.1	51.5	76.3	98.1	118.5	141.4
Standard deviation (nm)	3.5	4.2	5.8	2.1	2.4	2.2	1.5	2.0

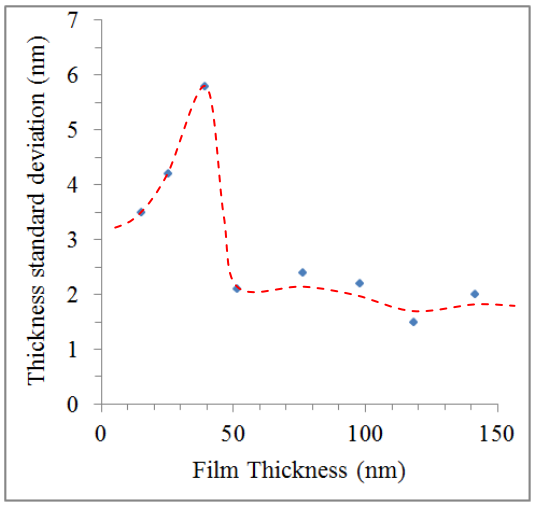

Figure 1: Thickness standard deviation of Al films sputtered on silicon. The standard deviations were obtained from the film surface profiles.

For sputtering aimed at producing films less than 10 nm thick, aluminum fragments were observed on the silicon surface with FE-SEM. They appear thin, scattered, and well separated, in addition their boundaries were irregular. The formation of clearly shaped

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nano-sized Al islands starts at aimed average film thicknesses of 10 nm, which defines the critical thickness for coalescence of sputtered Al nanoparticles. At Al film thickness of 15.4 nm, Al nanoparticles observed on top of Si surface are still isolated, see Figure 2 (a); their lateral size is within 10-30 nm . Similar observation has been reported for thermally evaporated Al thin film (8), where AFM images showed small nuclei for film thickness of 60 nm. At early stage deposition of evaporated Al, nucleation sites were formed, which then start to cumulate Al onto Si surface. Unlike sputtering, it is believed that that this process proceeds for longer time since Al atoms have less ability to overcome the Si surface diffusion barrier due to the low incident energy (9) and the low temperature of the sample; thus the coverage is reduced during this nucleation phase. Reduction of the coverage and low diffusion delay the nucleation phase. For sputtering, as the deposition of Al species proceeds, the size of the Al islands grow quicker due to the incident energy and to the buildup of a strong electric field around these nuclei.

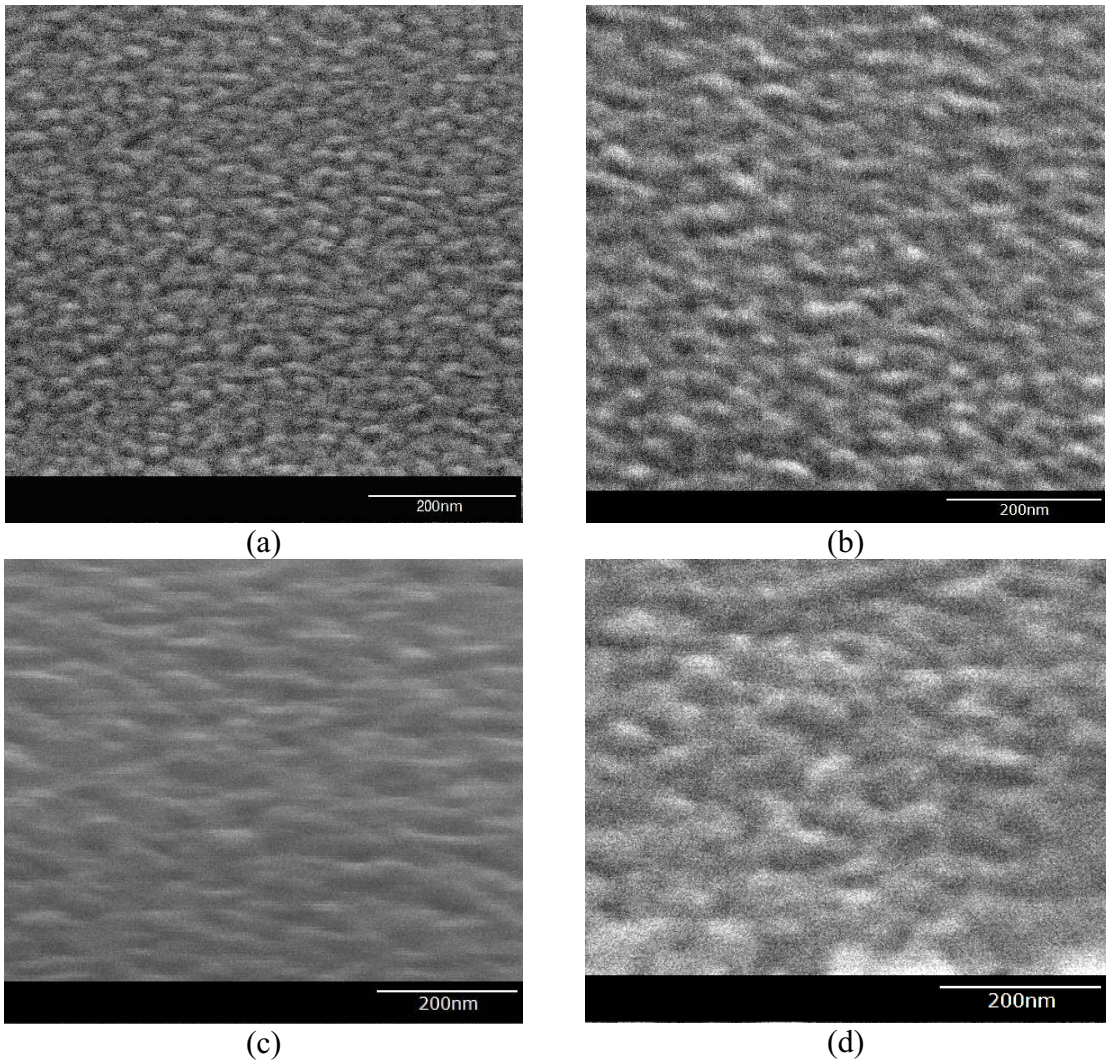

Figure 2: SEM top/cross-sectional view of Al film with thickness of (a) 15.4nm, (b) 39.1nm, (c) 76.3 nm, and (d) 118.8nm.

The coalescence between islands was observed in samples that had more sputtered Al, see Figure 2 (b). When further sputtering is done the gaps between islands start to be filled up with Al and grains merge into the adjacent ones, see Figure 2 (c). As a result, Al

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surface tends to smooth out and does more so as Al species are further sputtered on to the silicon substrate. This is in line with the observed reflectance increase and resistivity decrease of the same films, as will be discussed in the following sections. With further deposition, however, Al grains grow further as indicated by an increase of the surface roughness, see Figure 1 and Figure 2 (d).

In the depth profile of a 38nm thick Al film, see Figure 3, the surface roughness clearly appears higher than that of the initial silicon wafer. This suggests that Al nanoparticles cover the film surface as a result of a continuous involvement during the film growth of Al that have capability to grow further instead of spilling over the surface. Even in late stage, after island coalescence, the aluminum continues to attach to the deposited film in the form of sturdy clusters that can have (or grow to) nanometer size particles. Additionally, depth profiling of the deposited film showed that the formation of separated nanoparticles taller than 20nm is difficult to achieve by Al sputtering. This was attributed to particle coarsening, which causes the growth main component, i.e., along vertical direction, to compete with the lateral growth.

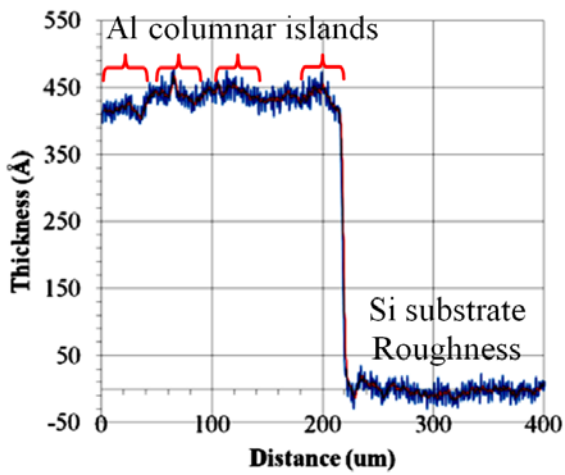

Figure 3: Depth profile of Al film thickness of ~40 nm. The Ag layer features (on average about 30nm wide) is larger than the Si substrate initial roughness.

The above depicted sequence for deposition of Al ultra-thin films gives a picture of the mechanisms of layer formation by sputtering from nucleation of islands to island growth up to formation of continuous films.

2) Reflectance of Ultra-Thin Aluminum Films

Figure 4 shows the reflectance of Al ultra-thin films, as a function of film thickness. The represented values are the average in visible light range from 400 to 800 nm for each Al film thickness, normalized by the highest reflectance value in that spectral range. As can be readily seen, the reflectance decreases to a minimum value at Al film thickness of 45 nm, and increases to the maximum at Al film thickness of 80 nm, and then decays gradually with the increase of thickness. Aside from the unique effects of Al band structure, the intensity of reflected light is strongly correlated to the structure to the ultra-thin Al film. During the sputter-deposition, the Al islands are initially formed on the Si surface. The small Al islands can cause the scattering of the incident beam, leading to the reduction of reflectance. As the deposition proceeds, a smoother surface is achieved due to the fact that the regions between islands get filled with Al clusters and nanoparticles.

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Throughout this process, adjacent islands merge into each other causing the total boundary surface to increase. As a consequence, the reflectance increases; it reaches its maximum at 80 nm, i.e., when the coalescence is completed. With further deposition, the reflectance continues to decrease slowly as a result of the increase of light trapping, in line with nanoscale porosity of the film surface seen in high resolution FE-SEM.

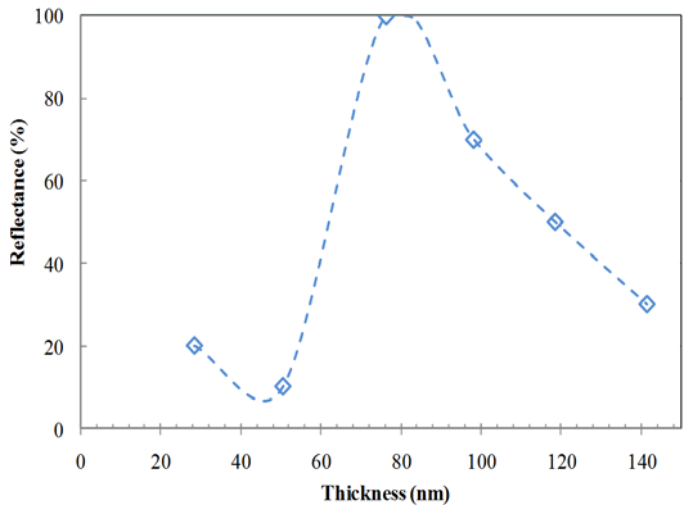

Figure 4: Trend of the Normalized Reflectance of Al films grown by sputter-deposition versus film thickness. Normalization was done relative to the maximum value.

3) Resistance of Ultra-Thin Aluminum Films

Nanostructured Al films electrical properties are affected by coverage reduction that leads to island formation and local microstructure variations (10, 11). Therefore, electrical properties are expected to contain information on the Al film formation phases. Resistivity of the Al films was calculated from their sheet resistance and the corresponding film thicknesses, obtained by averaging the depth profiles. Since the roughness is prominent in all samples it cannot be neglected in the film electric conduction.

As shown in Figure 5, the resistivity strongly depends on the Al film thickness. For samples with film thickness less than 40 nm, the resistivity increases dramatically when thickness decreases. Higher values are due to larger inter-island gaps, which disrupt the electric connectivity between islands. With the Al thickness in the range of 50 nm to 120 nm, the resistivity becomes nearly constant and attains its lowest value that is 6.1×10^{-8} Ω-m, the bulk resistivity. Several scenarios for the resistivity asymptotic behavior are possible: (i) most of the electric conduction is achieved through highly conductive island boundaries, (ii) as the Al film thickness increases, the film inner structure becomes almost continuous with a minimum density of holes and boundaries, or (iii) adjacent islands join without any intercalary oxide. TEM imaging and Z-contrast can contribute in clarifying the inter-islands material structure and composition.

During deposition of ultra-thin films less than 20nm (i.e., short sputtering time) coalescence process is not yet fully effective, thus Al nanoparticles are still disconnected. Non-classical conduction modes between those nano-particles (such as tunneling through barriers,…) are likely to occur, which enable current transmission to the islands, this would be statistically along the main direction of the applied electric field. In late stages of Al sputtering, the gaps between Al nanoparticles and islands close. Therefore, conventional conduction modes in the metal layer are established.

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The abrupt variation in the conduction mode between Sample 2 and Sample 3 seen in Figure 5 accounts for the observed sharp conductivity transition. Since it coincides with the variation in reflectance and standard deviation of surface topography, it is believed to be due to a structural change of the Al film.

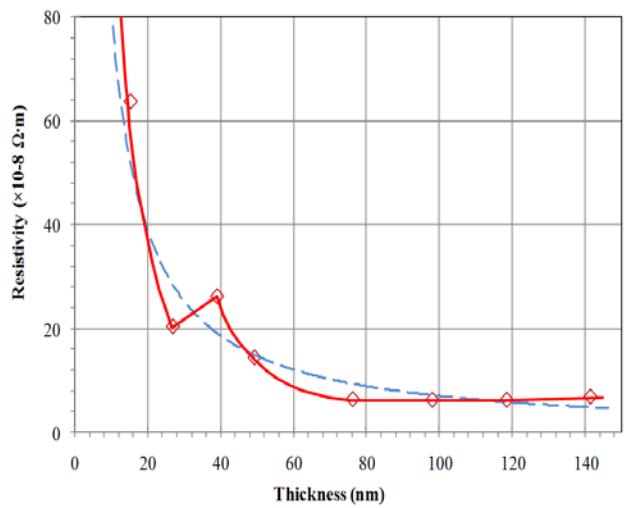

Figure 5: Electrical resistivity of sputter-deposited Al films vs. thickness. (Solid curve represents the actual variation, while the dashed line neglects the resistivity transition).

The strong correlation of the resistivity and reflectance data at the nanoscale level is seen at all regimes of our test samples. Both electrical and optical properties can be explained by the nearly continuous nanostructure of Al layers with thickness above critical thickness (12). The transition of these properties at around 40nm is also correlated to the structure transformation.

Conclusions

Sputter-deposition Al ultra-thin films on Si starts with the formation of isolated nanometer size islands, which then grow in the range of 10 to 60 nm, thereafter islands coalesce and inter-island voids and boundaries are progressively removed. The structure and surface topography of such Al films appeared well correlated with the film resistivity and reflectance. Al bulk resistivity is reached for films about 100 nm thick; it suggests that a continuum comprising metal grains is formed, the material structure is ultimate maintained in the submicron regime.

The resistivity and reflectance of the sputter-deposited Al showed abrupt transitions at around 40nm; such behavior is in line with observed structural changes. The detected high resistivity of ultra-thin films suggests that the material is mostly discontinuous. Non-classical electronic effects are likely to contribute to the inter-island electronic transport.

Finally, the formation of isolated Al nanoparticles can only be achieved at low Al sputtering regime, with the substrate being maintained at room temperature.

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